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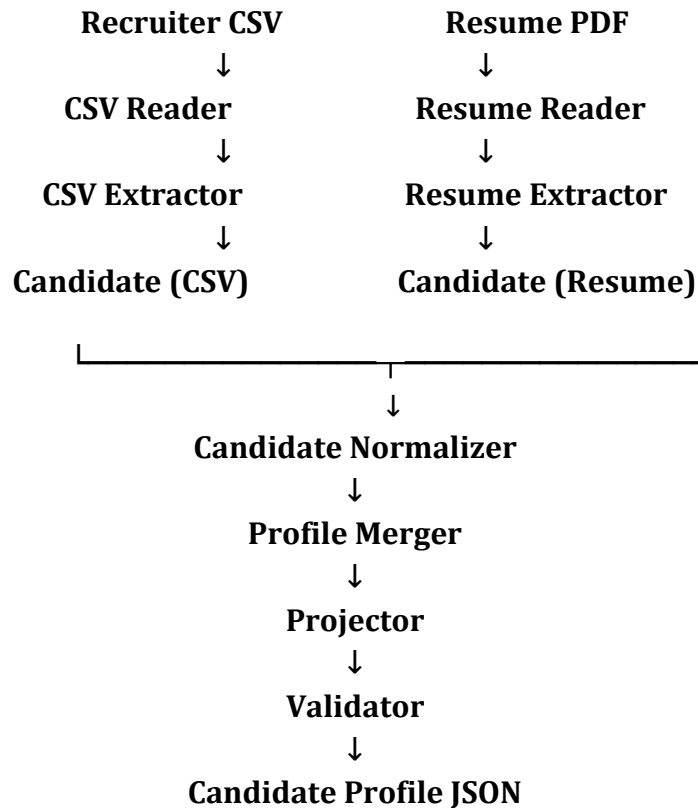
Candidate Profile Transformer

Design Overview

1. Problem Understanding

Recruiters often receive candidate information from multiple sources such as recruiter databases and resumes. Although both sources describe the same candidate, the information can be incomplete, inconsistent or duplicated. The objective of this project was to build a modular application that combines both sources into a single, clean candidate profile while keeping the solution simple, maintainable and easy to extend.

2. Overall Architecture



3. Solution Approach

Instead of implementing the complete workflow inside a single script, I separated the application into small independent modules. Each module performs one specific responsibility such as reading files, extracting information, normalizing values, merging candidate data or validating the final output. This design improves readability and makes future enhancements easier.

4. Module Responsibilities

Module	Responsibility
Readers	Read recruiter CSV and resume PDF.
Extractors	Extract information and create Candidate objects.
Normalizers	Normalize phone numbers, skills and dates.
Merger	Merge candidate information from multiple sources.
Projector	Generate configurable JSON output.
Validator	Validate mandatory fields before saving output.
Models	Represent the canonical candidate profile.

5. Key Design Decisions

I intentionally chose a lightweight rule-based implementation because the objective of the assignment was to demonstrate clean software engineering practices rather than build a production-scale NLP system. The application also preserves provenance and confidence metadata internally while exposing only the required fields in the final JSON.

6. Merge Strategy

- Recruiter data is preferred for candidate name, email and phone number.
- Resume data is preferred for skills, education and experience.
- Duplicate emails, phone numbers and skills are removed during merging.
- Confidence values are combined to calculate an overall confidence score.

7. Assumptions

- Each recruiter CSV contains information for one candidate.
- Resume contains recognizable section headings.
- Phone normalization assumes Indian phone numbers.
- Skill normalization is based on predefined mappings.

8. Current Limitations

The current implementation uses rule-based extraction, therefore resumes with significantly different formats may require additional parsing rules. Confidence values are heuristic and intended to demonstrate the overall design.

9. Future Scope

- Support additional structured data sources.
- Improve resume parsing using NLP.
- Enhance confidence scoring.
- Expose the pipeline as a REST API.
- Add unit and integration tests.

Conclusion

This project demonstrates a complete end-to-end pipeline that transforms structured and unstructured candidate information into a unified profile. The primary focus was to build a clean, modular and extensible solution that satisfies the assignment requirements while remaining easy to understand and maintain.